

Bachelor Thesis Midterm

A Multimodal XR Museum with vitrivr and Godot

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22 July 2026

Key terms

Term	Meaning	Role in this project
VR	Virtual Reality	The fully virtual museum environment experienced through a headset.
XR	Extended Reality	The umbrella term covering VR, augmented reality, and mixed reality.
OpenXR	Cross-platform XR API	A common interface for headsets, controllers, tracking, and XR runtimes.
CLIP	Contrastive Language–Image Pre-training	Maps text and images to comparable numerical embeddings.
HUD	Head-up display	Shows search progress and status messages within the user's view.

Starting point

PREDECESSOR

VIRTUE

- › dynamically arranged image galleries in VR
- › retrieval with spatial exploration
- › Cineast and Cottontail DB
- › image collections



THIS THESIS

Multimodal XR museum

- › vitrivr-engine and pgvector
- › images, videos, and 3D models
- › multiple headsets through OpenXR
- › similarity as an interaction

Three project goals

01

Retrieve

vitivr-engine and pgvector provide semantic search over multimodal embeddings.

02

Experience

The XR application presents retrieved images, videos, and 3D models as interactive exhibits.

03

Explore

Users trigger new similarity searches from exhibits.

System overview

01

VR application



Input and interaction

keyboard, controllers, hand tracking

Placement strategies

wall slots, 3D stands

02

.NET application



Use cases

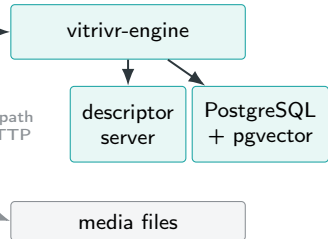
search, validation

Adapters

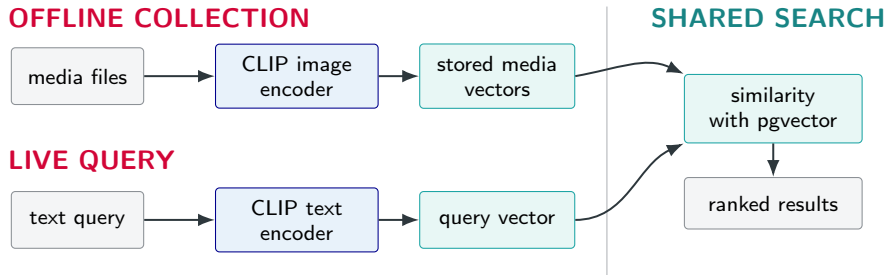
request, parser, loader

03

Retrieval + delivery

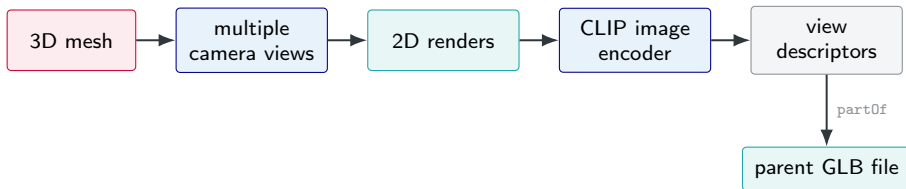


CLIP retrieval



Key distinction: the collection is encoded once, a live search only encodes the new query and compares it with the stored vectors.

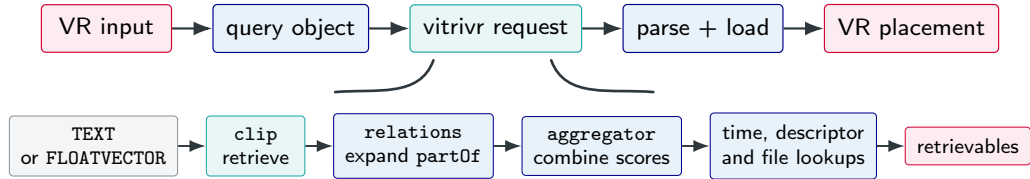
3D retrieval



Rather than encoding the mesh itself, the system searches over rendered views. A matching view is linked through `partOf` to the original GLB model, which is then loaded into the museum.

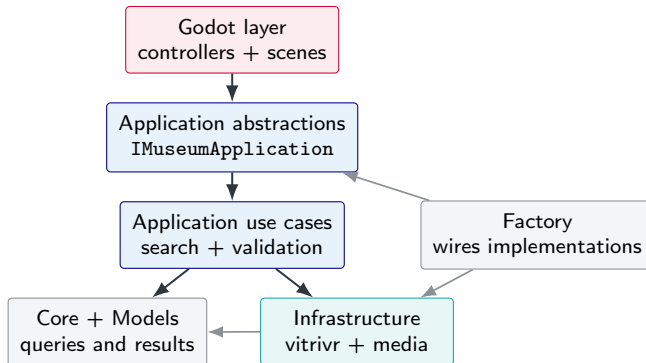
vitivr request

SIMPLIFIED REQUEST-RESPONSE FLOW



Only the input mapping changes. Response parsing, media loading, and VR placement are shared. The returned vector enables the next result-to-result search.

Godot and vitrivr remain replaceable



Why Godot?

01

Godot XR Tools

Ready-made building blocks for movement, interaction, grabbing, and XR interfaces.

02

OpenXR Vendors

Simple device integration for Quest and Focus; Windows uses the regular OpenXR runtime.

Controller and hand input use the same scenes and application logic.

Godot scene tree

Main



10 direct child nodes, grouped by role

XR RUNTIME

XrRenderProfile

StartXR

WorldEnvironment

STATE + WIRING

SearchSettingsStore

GameSettingsStore

SearchUseCaseFactory

PlatformSwitcher

SCENE INSTANCES

Player

XROrigin3D, camera + HUD
hands + controllers

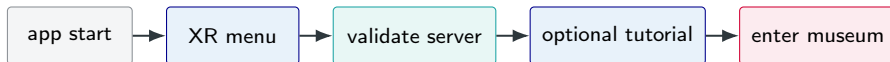
MenuNode

tutorial + keyboard
server + game settings

MuseumNode

search + room
media slots + 3D rooms

Entering the museum



`CanEnterMuseum` = server valid **and** (tutorial disabled **or** completed)

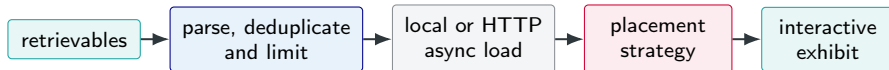
While in the menu

The museum is hidden, player movement is disabled, and configuration happens inside XR.

On transition

`PlatformSwitcher` moves the XR rig, swaps environment and HUD, and preserves museum state.

Media become exhibits



2D MEDIA

Images and videos use centred wall slots and aspect-aware frames.

3D MODELS

GLB objects are instanced on stands and fitted to their bounds.

Every exhibit stores its path, name, and CLIP vector for **Similar**. 3D models also support **Original Size**.

Current status

Backend & data

A1 Ingestion and extraction

Completed



A2 Similarity search

integration

Completed



A3 Performance

optimization

Partial



VR application

B1 Scene generation

Completed



B2 OpenXR setup

Completed



B3 User interaction

Partial



B4 Advanced user interaction

Partial



B5 Gesture-based interaction

Completed



Evaluation & documentation

D1 System evaluation

Partial



D2 Documentation

Missing



D3 User evaluation

Missing



Demo

Search → explore → inspect → search again

Evaluation questionnaire

01 SUS

10 standard statements about usability.

02 Project questions

Navigation, controls, responsiveness, 3D interaction, and design.

03 Context + feedback

VR experience, device, usage mode, and one open question.

Next steps

1. Reduce 3D asset sizes and resolve unsupported-material issues.
2. Integrate and test streaming to Apple Vision Pro.
3. Run user evaluations.
4. Complete the reproducible setup and documentation.
5. Finish the thesis.

TAKEAWAY

The prototype modernises the VIRTUE stack and already closes the loop from semantic query to spatial exploration.

Godot turns retrieval results into a device-aware XR experience.